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FACULTY OF ENGINEERING AND TECHNOLOGY
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CEF440: INTERNET PROGRAMMING AND MOBILE PROGRAMMING
DESIGN AND IMPLEMENTATION OF A MOBILE APP FOR COLLECTION OF USER
EXPERIENCE DATA FROM MOBILE NETWORK SUBSCRIBERS.

TASK 5: UI DESIGN AND IMPLEMENTATION

GROUP 2

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1. Introduction

This report documents Task 5 of the CEF 440 project: UI Design and Implementation for the QoETrack mobile application. Building directly on the system architecture established in Task 4 (System Modelling and Design), this task translates the UML-defined components including the Mobile Subscriber actor, the QoE Mobile App system boundary, and the seven identified use cases into a functional, visually branded Android user interface.

Task 5 is structured around three deliverables: App Identity (brand name, colour palette, typography, and logo), Visual Design (Figma wireframes and high-fidelity mockups for all seven screens), and Frontend Implementation (Android XML layouts and Kotlin activity controllers for each screen).

2. Relationship to Task 4 System Modelling and Design

The UI design and implementation in Task 5 is a direct realization of the system model produced in Task 4. Each screen implemented corresponds to a use case or process identified in the Task 4 UML diagrams, as summarised below.

Task 5 Screen	Task 4 UML Reference	Relationship
Splash Screen	Context Diagram QoE Mobile App	Entry point to the system; establishes app identity before subscriber interaction begins.
Onboarding Screen	Use Case: View Privacy Notice; Sequence Diagram 4	Introduces the app and its three core functions to new subscribers before permissions are requested.
Permission Screen	Use Case: Grant or Revoke Permissions; Sequence Diagram 3	Implements the permission grant/revoke flow where Android OS returns permission result to the app.
Home Dashboard	Use Case: Collect Network Metrics; DFD P1 Metric Collector	Displays live dBm, network type, latency and GPS from TelephonyManager and LocationManager.
Feedback Form	Use Case: Submit Satisfaction Rating; Sequence Diagram 1	Implements the periodic feedback prompt → rating submit → metric snapshot → INSERT to SQLite flow.
Settings Screen	Use Cases: Configure Prompt Frequency, Request Data Deletion	Allows subscribers to set collection interval and trigger the DELETE flow from Sequence Diagram 5.
Privacy Notice Screen	Use Case: View Privacy Notice; Sequence Diagram 4	Renders the full privacy statement as referenced in the Task 4 DFD and Use Case Diagram.

3. App Identity

3.1 App Name and Tagline

The application was named QoETrack a compound of QoE (Quality of Experience) and Track, reflecting the app's core purpose of continuously tracking and reporting subscriber network experience. The tagline chosen is: "Rate It. Report It. Improve It." a three-part phrase that communicates the subscriber's role, the data collection function, and the intended benefit to network quality.

3.2 Colour Palette

The following brand colours were defined and applied consistently across all screens:

Role	Hex Code	Usage
Primary (Navy)	#1A3A6B	Top bar, signal hero card, buttons, headings, nav active state
Accent (Teal)	#00C9A7	Logo circle, step circles, metric underlines, activity dots, badge fills
Background	#F4F8FC	Screen background, card surface variant
Surface	#FFFFFF	Card backgrounds, activity rows, feature rows
Text Primary	#1C2B3A	Body text, card titles, section headings
Text Muted	#607080	Captions, subtitles, secondary labels
Error/Danger	#E53935	Request Data Deletion row in Settings

3.3 Typography

Two typefaces were selected for the application:

- Poppins (Bold, SemiBold, Regular) used for all headings, the app name, button labels, card titles, and metric values. Poppins provides a modern, geometric feel appropriate for a data-driven application.
- Inter (Regular) used for body text, captions, timestamps, and supporting labels. Inter is optimised for screen legibility at small sizes.

3.4 Logo

The logo consists of a circular teal shape (#00C9A7) containing the text "QoET" in white Poppins Bold, paired with the full app name "QoETRACK" in large white bold text beneath it. This combination is used on the Splash Screen. The circular form references signal and connectivity, and the teal colour distinguishes the brand mark against the dark navy background.

4. Visual Design Figma Mockups

4.1 Design Tool and Setup

All screen mockups were designed in Figma using an Android Small frame (360 × 800px). Color styles, text styles, and reusable components were defined at the start of the project and applied consistently across all seven screens. A prototype flow was configured to link screens together, enabling a clickable walkthrough from Splash through to the Privacy Notice.

4.2 Screen Designs

Screen 1 Splash Screen

The Splash Screen features the navy gradient background (#1A3A6B to #0D2850), the circular teal logo centred vertically at 42% of screen height, the app name in 34sp bold white text, the tagline "Rate It. Report It. Improve It." in muted blue (#7AABCF), and an animated teal loading bar at the bottom of the screen. The screen auto-navigates after 2.6 seconds.

Screen 2 Onboarding Screen

The Onboarding Screen is divided into two sections. The top 40% is a dark navy panel containing the app name and three teal icon boxes representing Signal () , Location (), and Feedback (). The lower 60% presents a scrollable white area with the title "How QoETRACK Works", three numbered feature rows with teal circle step indicators and descriptive text, and a full-width navy Get Started button at the bottom.

Screen 3 Permission Request Screen

The Permission Screen presents two permission cards (Location Access and Network Access) with toggle switches, an informational note, and an Allow & Continue button. A Skip for now text link is provided below the primary button.

Screen 4 Home Dashboard Screen

The Dashboard is the primary screen of the application. It features a navy top bar with the app name and a live sync indicator, a large navy signal hero card displaying the current dBm value in teal 48sp text with network type and quality badge below, three horizontal metric mini-cards (Network, Latency, Location), a Recent Activity section with three timestamped rows, a Sync Now button, and a bottom navigation bar with four tabs.

Screen 5 Feedback Form Screen

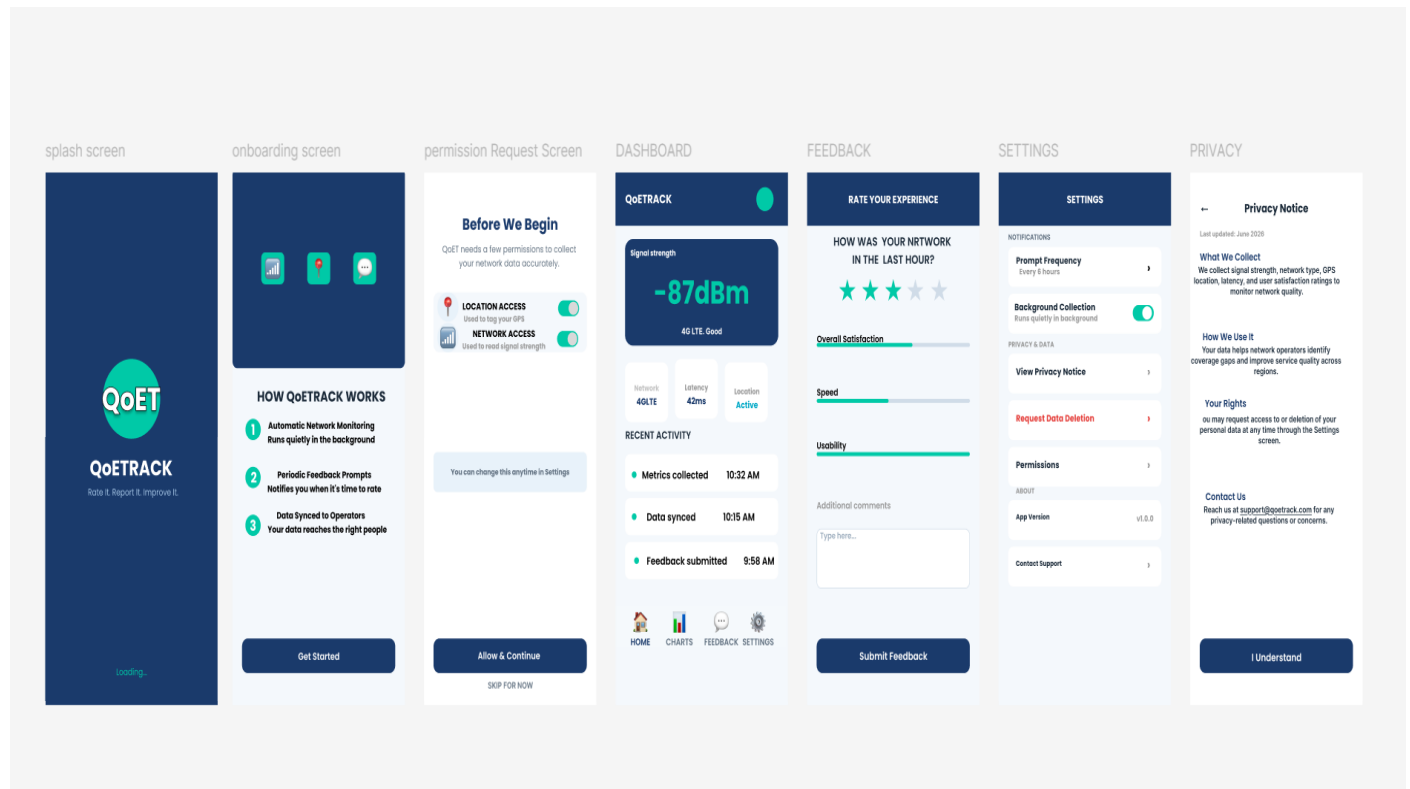
The Feedback Screen presents a 5-star rating row, three labelled sliders for Overall Satisfaction, Speed, and Usability, an optional comments text field, and a Submit Feedback button. The screen is triggered periodically by the background service as defined in Sequence Diagram 1 from Task 4.

Screen 6 Settings Screen

The Settings Screen is organized into three labelled sections: Notifications (Prompt Frequency, Background Collection toggle), Privacy & Data (View Privacy Notice, Request Data Deletion in red, Permissions), and About (App Version, Contact Support).

Screen 7 Privacy Notice Screen

The Privacy Notice Screen displays the full data collection policy in formatted sections: What We Collect, How We Use It, Your Rights, and Contact Us. An I Understand button is fixed at the bottom of the screen.



5. Frontend Implementation

5.1 Technology Stack

The frontend was implemented using Android Studio with Kotlin as the programming language and XML-based view layouts. The Material Components library was used for theming and component styling. ConstraintLayout and LinearLayout were used as the primary layout containers depending on the complexity of each screen.

Component	Detail
Language	Kotlin
Layout System	Android XML (ConstraintLayout, LinearLayout, FrameLayout, ScrollView)
Minimum SDK	API 24 (Android 7.0)
Target SDK	API 37 (Android 15)
Theme	Theme.AppCompat.Light.NoActionBar
Typography	Poppins (Bold, SemiBold, Regular) + Inter via res/font/
Dependencies	constraintlayout:2.2.1, cardview:1.0.0, material:1.14.0, appcompat:1.7.1

5.2 Implemented Screens

5.2.1 Splash Screen SplashActivity.kt + activity_splash.xml

The Splash Screen is implemented as a LinearLayout with a navy gradient background. The logo FrameLayout uses an oval drawable (shape_circle_teal.xml) to produce the circular teal shape. The loading bar is implemented as a FrameLayout containing a track View and a fill View whose width is animated using a ValueAnimator from 0 to the full track width over 1.3 seconds.

SplashActivity.kt manages a five-stage animation sequence using ObjectAnimator and OvershootInterpolator: (1) logo bounces in with scale $0.2 \rightarrow 1.0$, (2) app name slides up from 25dp, (3) tagline fades in, (4) loading bar fills, (5) navigation fires after 2.6 seconds. SharedPreferences is used to detect first launch if onboarding_complete is false, the app navigates to OnboardingActivity; otherwise it goes directly to DashboardActivity.

5.2.2 Onboarding Screen OnboardingActivity.kt + activity_onboarding.xml

The Onboarding Screen uses a vertical LinearLayout root with layout_weight to split the screen 40/60 between the dark banner and the scroll content area. This was a critical design decision: an earlier implementation using ConstraintLayout with app:layout_constraintHeight_percent caused the screen to fail to render. LinearLayout with layout_weight is the correct approach for proportional height division on Android.

OnboardingActivity.kt runs a staggered entrance animation using `Handler.postDelayed`, revealing the top section, title, and each feature row at 150ms intervals. The Get Started button saves the `onboarding_complete` flag to `SharedPreferences` and navigates to `DashboardActivity`.

5.2.3 Dashboard Screen `DashboardActivity.kt` + `activity_dashboard.xml`

The Dashboard is the most complex screen. It uses a `ConstraintLayout` root containing a fixed top bar, a `ScrollView` content area, and a fixed bottom navigation bar. The signal hero card uses a gradient drawable (`bg_signal_card.xml`) applied to a `CardView`. The three metric mini-cards are arranged in a horizontal `LinearLayout` with equal weight.

`DashboardActivity.kt` contains the `updateSignal()` method which accepts a dBm integer, network type string, and latency integer, and updates all relevant UI elements including the colour of the dBm value (teal for Good/Excellent, amber for Fair, red for Poor). A simulated live update cycle fires every 5 seconds using `Handler` to demonstrate the dynamic data flow.

5.3 XML Drawable Resources

The following custom drawable files were created to implement the visual design:

Drawable File	Purpose
<code>shape_circle_teal.xml</code>	Oval shape for logo circle and step number circles
<code>shape_loading_fill.xml</code>	Teal rectangle for animated loading bar fill
<code>shape_loading_track.xml</code>	Dark rectangle for loading bar background track
<code>shape_btn_primary.xml</code>	Navy rectangle with 14dp corner radius for all primary buttons
<code>shape_step_circle.xml</code>	Oval shape for numbered step indicators on onboarding rows
<code>shape_icon_box.xml</code>	Teal rounded rectangle for the three icon boxes in onboarding banner
<code>bg_signal_card.xml</code>	135° gradient (navy to dark blue) for the dashboard signal hero card
<code>bg_metric_card.xml</code>	White rectangle with divider-coloured stroke for metric mini-cards
<code>bg_badge_good.xml</code>	Semi-transparent teal rectangle for signal quality badge
<code>dot_accent.xml</code>	10×10dp teal oval for Recent Activity list indicators

5.4 Navigation and State Management

Navigation between screens is handled through explicit `Android Intent` objects. The application consists of seven screens: Splash, Onboarding, Permission, Dashboard, Feedback, Settings, and Privacy Notice.

On first launch, the application follows a linear onboarding flow: Splash → Onboarding → Permission → Dashboard

On all subsequent launches, the onboarding and permission screens are skipped, resulting in a shortened flow: Splash → Dashboard

From the Dashboard, the user can navigate to three secondary screens via the bottom navigation bar:

- Dashboard → Feedback (via the Charts/Feedback nav button)
- Dashboard → Settings (via the settings nav button)
- Settings → Privacy Notice (via the View Privacy Notice row)
- Settings → Permission (via the Permissions row)

Both `SplashActivity` and `OnboardingActivity` declare `android:noHistory="true"` in the `AndroidManifest.xml`, removing them from the back stack immediately upon navigation. This prevents the user from returning to either screen using the device back button once they have been dismissed.

A boolean flag named `onboarding_complete` is persisted in a `SharedPreferences` file identified by the key `goetrack_prefs`. This flag is set to `true` when the user taps Get Started on the Onboarding screen. On every subsequent app launch, `SplashActivity` reads this flag during `onCreate()` to determine the appropriate navigation destination forwarding first-time users through the onboarding flow and returning users directly to the Dashboard.

6. Implementation Challenges and Resolutions

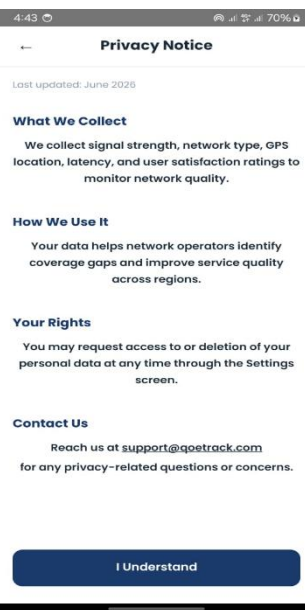
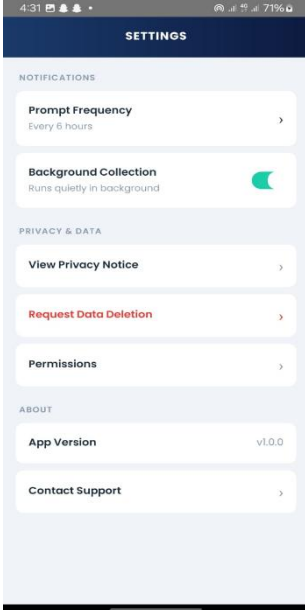
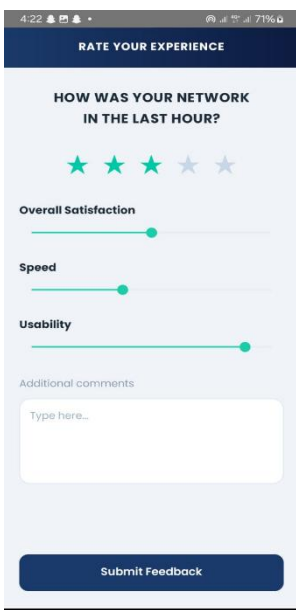
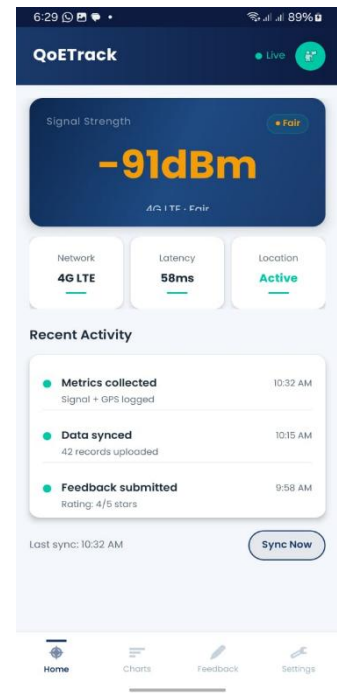
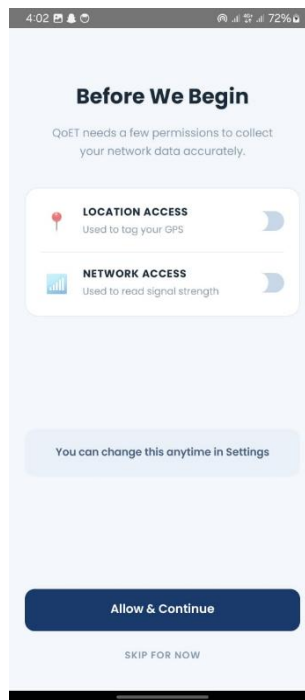
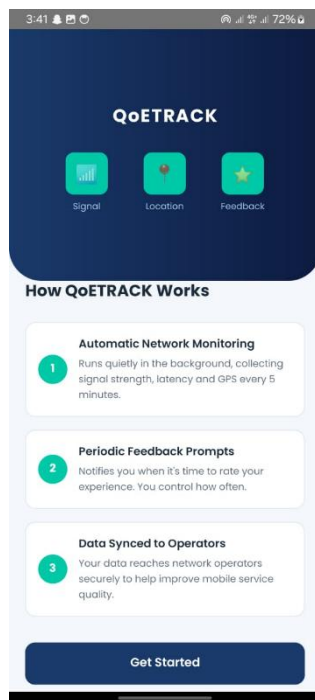
Here's the adjusted and expanded challenges table reflecting your actual build process:

Challenges and Resolutions

- Gradle IllegalDependencyNotation Groovy-style dependency strings used in a Kotlin DSL project. Fixed by replacing with full implementation("group:artifact:version") format.
- compileSdk android-35.1 not found Invalid compileSdk {version = release (35)} block syntax generated a non-existent sub-version. Fixed by replacing with compileSdk = 35.
- AAR dependency compatibility error Dependencies required compileSdk 35 or later but project was set to 34. Fixed by bumping both compileSdk and targetSdk to 35.
- android:hintTextColor resource linking failed Not a valid XML attribute on EditText. Removed from layout and applied in code via setHintTextColor().
- android:paintFlags resource linking failed Not a valid XML layout attribute. Removed from layout and applied in code via paintFlags or Paint.UNDERLINE_TEXT_FLAG.
- Onboarding screen not displaying onboarding_complete flag already true from a previous install, causing SplashActivity to skip onboarding silently. Fixed by uninstalling the app to reset SharedPreferences.
- OnboardingActivity not navigating correctly withEndAction {PermissionActivity() } only instantiated the class without triggering navigation. Fixed by replacing with withEndAction { navigateToPermission() }.
- Screen orientation warnings android:screenOrientation="portrait" deprecated in Android 16. Removed from all activity declarations in AndroidManifest.xml.
- Duplicate core-ktx dependency Already declared by the project template; added again manually. Removed the duplicate entry.
- Logo rendering as square Background drawable used android:shape="rectangle". Fixed by changing to android:shape="oval".
- Font references failing res/font/ folder missing and font files not renamed correctly. Fixed by creating the directory, downloading Poppins from Google Fonts, and renaming all three weight files correctly.

7. Screens Summary

Screen	Layout File	Activity File	Status
Splash	activity_splash.xml	SplashActivity.kt	Implemented & tested
Onboarding	activity_onboarding.xml	OnboardingActivity.kt	Implemented & tested
Permission	activity_permission.xml	PermissionActivity.kt	Implemented & tested
Dashboard	activity_dashboard.xml	DashboardActivity.kt	Implemented & tested
Feedback Form	activity_feedback.xml	FeedbackActivity.kt	Implemented & tested
Settings	activity_settings.xml	SettingsActivity.kt	Implemented & tested
Privacy Notice	activity_privacy.xml	PrivacyActivity.kt	Implemented & tested



8. Conclusion

Task 5 successfully translates the system model from Task 4 into a functional, visually branded Android application. The QoETrack identity built around the navy and teal palette, Poppins typography, and the circular logo provides a professional and cohesive visual language consistent across all screens.

All seven screens were fully implemented and tested on a physical Android device. The complete application flow from Splash through Onboarding, Permission Request, Dashboard, Feedback Form, Settings, and Privacy Notice operates as designed, with all navigation transitions, animations, UI components, and brand styling functioning correctly. The QoETrack application represents a complete frontend implementation of the system architecture defined in Task 4. The implementation process surfaced and resolved several technical challenges related to Gradle build configuration, SDK version management, layout system constraints, and drawable definitions. Each challenge was diagnosed and resolved systematically, with the solutions documented in Section 6 of this report.